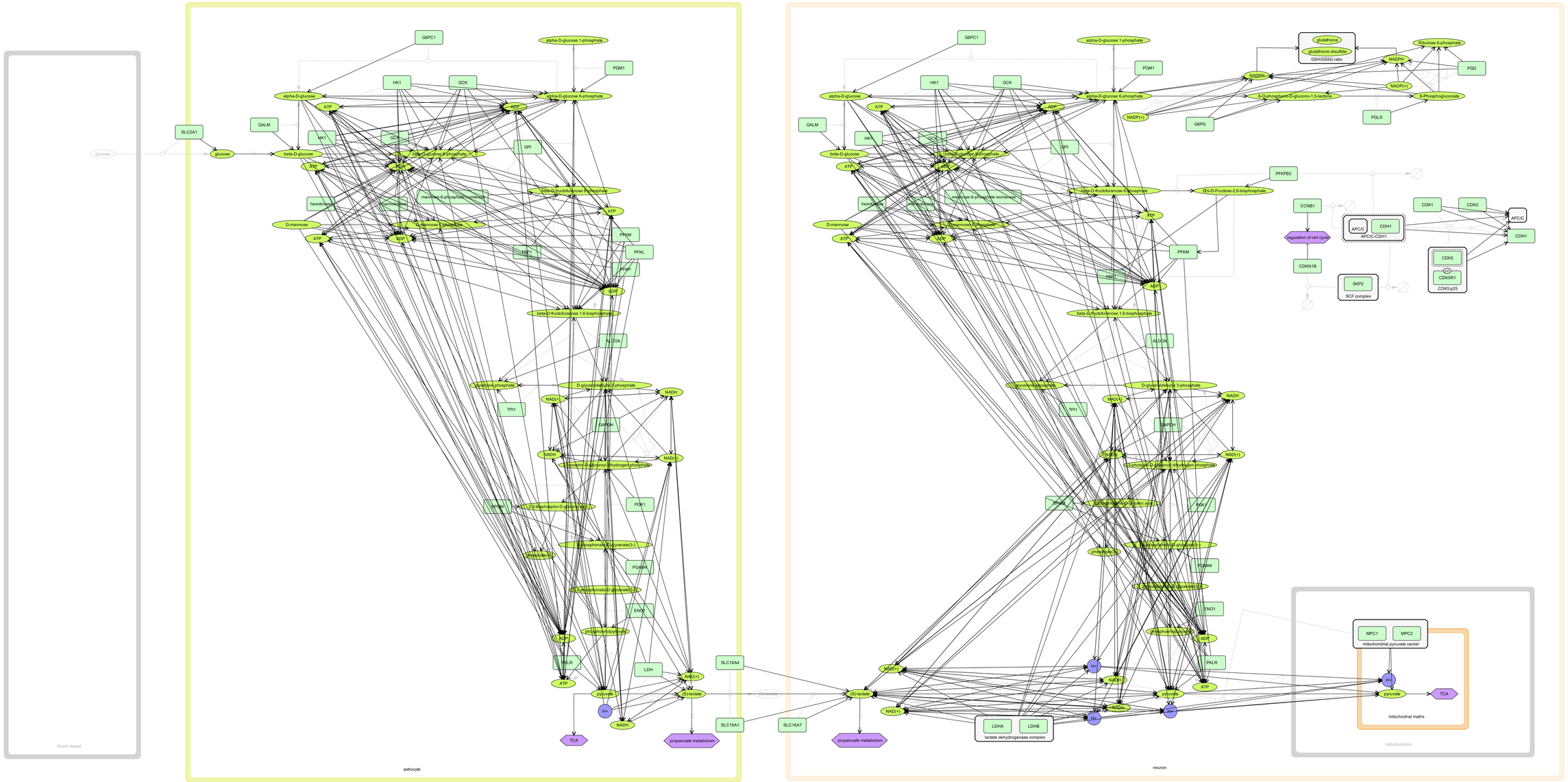

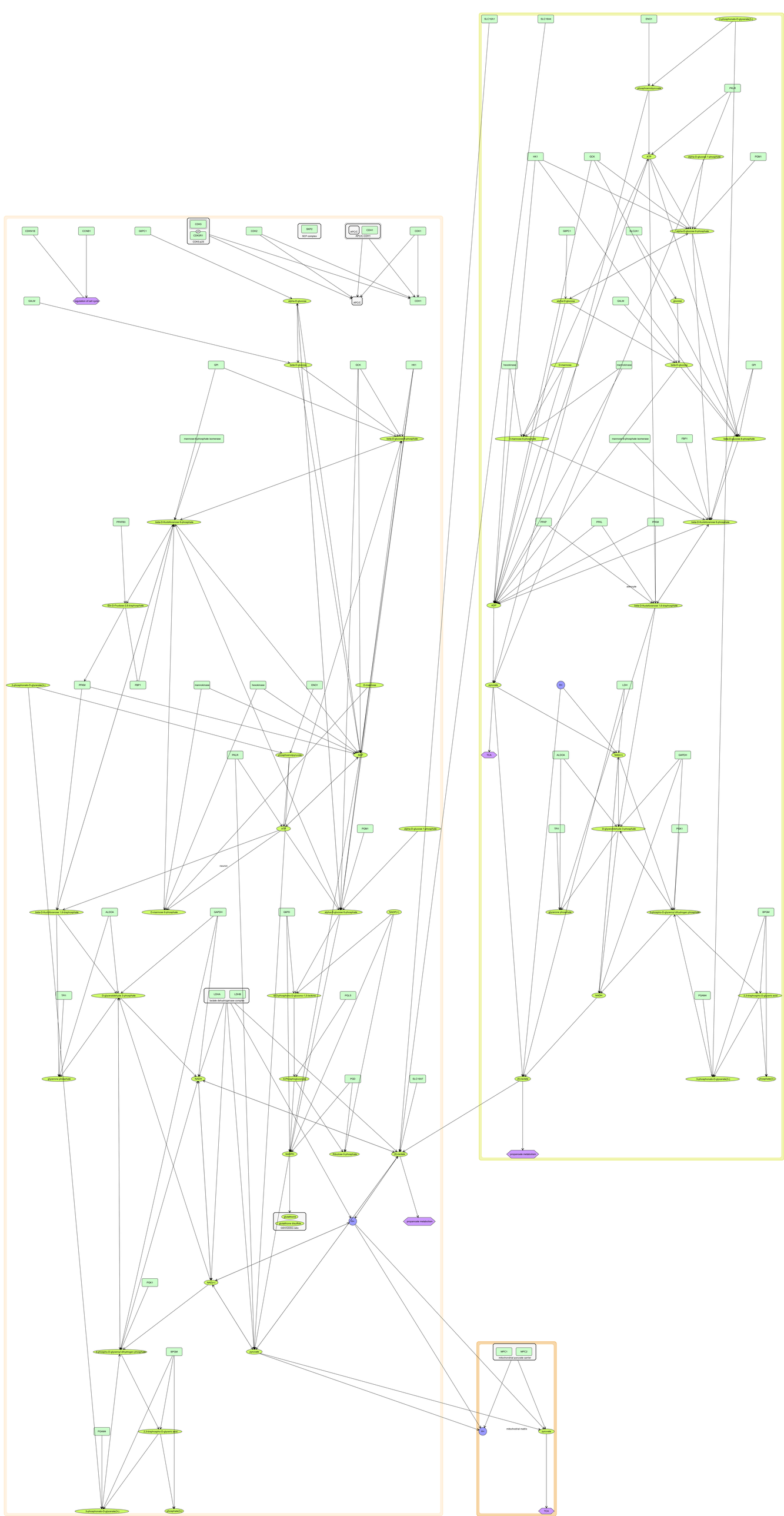

blood vessel
glucose
SLC2A1
glucose
G6PC1
alpha-D-glucose 1-phosphate
PGM1
HK1
GCK
alpha-D-glucose
ATP
ADP
alpha-D-glucose 6-phosphate
GALM
HK1
GCK
beta-D-glucose
beta-D-glucose 6-phosphate
GPI
ATP
ADP
beta-D-fructofuranose 6-phosphate
mannose-6-phosphate isomerase
hexokinase
mannokinase
D-mannose
D-mannose 6-phosphate
ATP
ADP
ATP
PFKM
FBP1
PFKL
PFKP
ADP
beta-D-fructofuranose 1,6-bisphosphate
ALDOA
glycerone phosphate
D-glyceraldehyde 3-phosphate
NAD(+)
NADH
TPI1
GAPDH
NADH
NAD(+)
3-phospho-D-glyceroyl dihydrogen phosphate
BPGM
2,3-bisphospho-D-glyceric acid
PGK1
3-phosphonato-D-glycerate(3-)
phosphate(3-)
PGAM4
2-phosphonato-D-glycerate(3-)
ENO1
phosphoenolpyruvate
ADP
PKLR
LDH
ATP
NAD(+)
pyruvate
(S)-lactate
H+
NADH
TCA
propanoate metabolism
astrocyte
SLC16A4
SLC16A1
(S)-lactate
SLC16A7
G6PC1
alpha-D-glucose 1-phosphate
PGM1
HK1
GCK
alpha-D-glucose
ATP
ADP
alpha-D-glucose 6-phosphate
GALM
HK1
GCK
beta-D-glucose
beta-D-glucose 6-phosphate
GPI
ATP
ADP
beta-D-fructofuranose 6-phosphate
mannose-6-phosphate isomerase
hexokinase
mannokinase
D-mannose
D-mannose 6-phosphate
ATP
ADP
ATP
NADPH
NADP(+)
G6PD
6-O-phosphono-D-glucono-1,5-lactone
glutathione
glutathione disulfide
GSH/GSSG ratio
NADPH
NADP(+)
Ribulose-5-phosphate
PGD
6-Phosphogluconate
PGLS
PFKFB3
Œ≤-D-Fructose-2,6-bisphosphate
PFKM
FBP1
ADP
beta-D-fructofuranose 1,6-bisphosphate
ALDOA
CCNB1
regulation of cell cycle
CDKN1B
APC/C
CDH1
APC/C-CDH1
SKP2
SCF complex
CDK1
CDK2
APC/C
CDH1
CDK5
p25
CDK5R1
CDK5:p25
glycerone phosphate
D-glyceraldehyde 3-phosphate
NAD(+)
NADH
TPI1
GAPDH
NADH
NAD(+)
3-phospho-D-glyceroyl dihydrogen phosphate
BPGM
2,3-bisphospho-D-glyceric acid
PGK1
3-phosphonato-D-glycerate(3-)
phosphate(3-)
PGAM4
2-phosphonato-D-glycerate(3-)
ENO1
phosphoenolpyruvate
ADP
PKLR
H+
NADH
ATP
pyruvate
NADH
H+
H+
NAD(+)
(S)-lactate
NAD(+)
LDHA
LDHB
lactate dehydrogenase complex
propanoate metabolism
MPC1
MPC2
mitochondrial pyruvate carrier
H+
pyruvate
TCA
mitochodrial matrix
mitochondrion
neuron

SLC16A1
SLC16A4
ENO1
2-phosphonato-D-glycerate(3-)
phosphoenolpyruvate
PKLR
HK1
GCK
ATP
alpha-D-glucose 1-phosphate
PGM1
CDKN1B
CCNB1
G6PC1
CDK5
p25
CDK5R1
CDK5:p25
CDK2
SKP2
SCF complex
APC/C
CDH1
APC/C-CDH1
CDK1
G6PC1
SLC2A1
alpha-D-glucose 6-phosphate
GALM
regulation of cell cycle
alpha-D-glucose
APC/C
CDH1
alpha-D-glucose
GALM
glucose
GPI
beta-D-glucose
GCK
HK1
hexokinase
D-mannose
mannokinase
beta-D-glucose
GPI
mannose-6-phosphate isomerase
beta-D-glucose 6-phosphate
D-mannose 6-phosphate
mannose-6-phosphate isomerase
FBP1
beta-D-glucose 6-phosphate
PFKFB3
beta-D-fructofuranose 6-phosphate
PFKP
PFKL
PFKM
beta-D-fructofuranose 6-phosphate
Œs-D-Fructose-2,6-bisphosphate
astrocyte
ADP
beta-D-fructofuranose 1,6-bisphosphate
2-phosphonato-D-glycerate(3-)
PFKM
FBP1
mannokinase
hexokinase
ENO1
D-mannose
pyruvate
H+
LDH
PKLR
phosphoenolpyruvate
ADP
TCA
ALDOA
NAD(+)
GAPDH
ATP
PGM1
alpha-D-glucose 1-phosphate
TPI1
D-glyceraldehyde 3-phosphate
PGK1
neuron
beta-D-fructofuranose 1,6-bisphosphate
ALDOA
D-mannose 6-phosphate
GAPDH
G6PD
alpha-D-glucose 6-phosphate
NADP(+)
glycerone phosphate
3-phospho-D-glyceroyl dihydrogen phosphate
BPGM
TPI1
D-glyceraldehyde 3-phosphate
LDHA
LDHB
lactate dehydrogenase complex
6-O-phosphono-D-glucono-1,5-lactone
PGLS
NADH
PGAM4
2,3-bisphospho-D-glyceric acid
glycerone phosphate
NADH
6-Phosphogluconate
PGD
SLC16A7
(S)-lactate
3-phosphonato-D-glycerate(3-)
phosphate(3-)
NADPH
Ribulose-5-phosphate
(S)-lactate
propanoate metabolism
glutathione
glutathione disulfide
GSH/GSSG ratio
H+
propanoate metabolism
PGK1
NAD(+)
3-phospho-D-glyceroyl dihydrogen phosphate
pyruvate
BPGM
MPC1
MPC2
mitochondrial pyruvate carrier
H+
mitochodrial matrix
pyruvate
PGAM4
2,3-bisphospho-D-glyceric acid
3-phosphonato-D-glycerate(3-)
phosphate(3-)
TCA